Dual Active Bridge Converter-II

Introduction

In the rapidly evolving landscape of electric vehicle (EV) charging infrastructure, efficient and bidirectional power conversion is paramount. The Dual Active Bridge (DAB) converter stands out as a critical component, offering high power density, galvanic isolation, and the ability to control power flow in both directions. This lecture delves deeper into the operational intricacies of DAB converters, building upon the foundational understanding of their two active full bridges connected via a transformer and series inductor. We explore various modulation techniques that enable precise control over power transfer, analyze the behavior of inductor currents under different operating conditions, and derive key relationships for power transfer and component selection. Understanding these concepts is essential for designing robust and optimized power converters for modern EV charging systems.

Core Concepts

Term	Definition	Significance
`Dual Active Bridge (DAB) Converter`	A power converter comprising two active full bridges connected by a high-frequency transformer and a series inductor, allowing bidirectional power flow.	Enables efficient, isolated, and bidirectional power transfer, crucial for EV charging and grid integration.
`Single Phase Shift (SPS) Modulation`	A control strategy where power flow is regulated by introducing a phase shift (delta) between the switching instances of the primary and secondary full bridges.	Simplest modulation, offering fundamental control over power direction and magnitude, but can lead to high RMS currents.
`Delta Star (δ*)`	The normalized phase shift, typically expressed as delta divided by 180 degrees, used in power transfer equations.	Simplifies calculations for power transfer and helps identify the maximum power point at δ* = 0.5 (delta = 90°).
`Inductance L`	The value of the series inductor placed between the	Crucial for controlling the current waveform, power

Term	Definition	Significance
	transformer and the full bridges.	transfer capability, and limiting current ripples; its selection impacts efficiency and size.
`RMS Current`	The root mean square value of the current flowing through components like the inductor, transformer windings, and switches.	Directly correlates with conduction losses and thermal stress in components; minimizing RMS current is a key design objective for efficiency.
`Backward Power Flow`	The ability of the DAB converter to transfer power from the secondary side (e.g., EV battery) back to the primary side (e.g., grid).	Essential for vehicle-to-grid (V2G) applications and regenerative braking, enhancing energy efficiency and grid stability.
`Resonant DAB Converters`	A variant of DAB converters that incorporates resonant networks (L-C circuits) to shape the voltage waveforms into sinusoidal forms.	Reduces switching losses by achieving soft switching and filters out higher harmonics, improving efficiency and electromagnetic compatibility.
`Multi-level DAB Converters`	DAB converters that utilize multi-level half-bridges instead of traditional two-level half-bridges.	Allows for more voltage levels, leading to reduced voltage stress on switches, lower RMS currents, and improved power quality.

Detailed Analysis

1. Modulation Schemes in DAB Converters

The Dual Active Bridge converter offers several modulation strategies to control power flow, each with its own characteristics regarding complexity, efficiency, and current ripple.

- **Single Phase Shift (SPS) Modulation:**
- This is the simplest and most common modulation technique.

- Power transfer is controlled by a single phase shift, `delta`, introduced between the switching instances of the primary (S1-S4) and secondary (S5-S8) full bridges.
- Specifically, the turn-on/off instances of S5 are shifted by `delta` relative to S1, S7 relative to S3, S6 relative to S2, and S8 relative to S4.
- All switches operate with a 50% duty cycle (turned on for half the switching cycle period).
- By varying `delta`, the amount and direction of power transferred between the input (A1B1) and output (A2B2) terminals can be precisely controlled. A leading phase shift typically results in power flow from A1B1 to A2B2, while a lagging phase shift enables backward power flow.
- **Dual Phase Shift (DPS) Modulation:**
- Involves two independent phase shifts: `delta` (between bridges) and `theta_1` (within the primary bridge) or `theta_2` (within the secondary bridge).
- `theta_1` controls the phase shift between diagonal switches in the primary full bridge (e.g., S1 and S4), while `theta_2` does the same for the secondary bridge (e.g., S5 and S8).
- This allows for more degrees of freedom in shaping the voltage waveforms and inductor current, potentially reducing RMS currents and improving efficiency.
- **Triple Phase Shift (TPS) Modulation:**
- Utilizes three independent phase shifts: `delta`, `theta_1`, and `theta_2`.
- This provides the highest degree of control over the converter's operation, enabling optimization for various parameters like efficiency, soft-switching range, and reduced RMS currents.
- TPS modulation introduces "zero states" in the voltage waveforms (VA1B1 and VA2B2), where the voltage across the bridge output is zero for a certain duration. This further shapes the inductor current.
- **Variable Frequency Modulation:**
- Can be combined with any of the phase shift modulation schemes (e.g., DPS with variable frequency, TPS with variable frequency).
- By changing the switching frequency (`FSW`), the impedance offered by the series inductor changes, which directly influences the amount of current flowing and thus the power transfer.
- This adds another layer of control, allowing for wider operating ranges or finer tuning of power.

2. Inductor Current Waveforms and Power Transfer Analysis

The behavior of the inductor current (`IL`) is central to understanding DAB converter operation and power transfer. The lecture details the current waveform for the Single Phase Shift (SPS) modulation, particularly when `V_in` is greater than `V_out/N` (where N is the turns ratio).

- **Inductor Current Waveform (V_in > V_out/N):**
- The current waveform is piecewise linear, characterized by different slopes determined by the voltage across the inductor (`V_L = V_in - V_out/N`).
- During different intervals within a half-switching cycle (`TS/2`), the slope of `IL` changes:
- `V_in + V_out/N / L` (rising sharply)
- `V_in - V_out/N / L` (rising slowly)
- `-V_in - V_out/N / L` (falling sharply)
- `-V_in + V_out/N / L` (falling slowly)
- The waveform repeats itself after `TS/2` . Key current points are defined as `I0` , `I1` , `I2` , `I3` , `I4` , with relationships like `I0 = -I2 = I4` and `I1 = -I3` .
- **Derivation of Power Transfer:**
- The average input current (`I_in_average`) is found by averaging the inductor current (`IL`) over a `TS/2` period. This involves calculating the area under the `IL` waveform (composed of triangles and trapezoids) and dividing by `TS/2` .
- After substituting the expressions for `I1` , `I2` , `T1` , and `T2` (derived from the slopes and time intervals), the `I_in_average` is given by: `I_in_average = (1 - δ*) * 2 * δ* * V_in / (2 * N * L * FSW)` where `δ*` is `delta / 180°` and `FSW = 1/TS` .
- The average input power (`P_in_average`) is then `V_in * I_in_average` : `P_in_average = V_in^2 * δ* * (1 - δ*) / (2 * N * L * FSW)`
- Assuming power balance (`P_out_average = P_in_average`), this equation represents the average power transferred.
- **Maximum Power Transfer:**
- To find the maximum power (`P_average_maximum`), the derivative of `P_average` with respect to `δ*` is set to zero.
- This yields `1 - 2δ* = 0` , implying `δ* = 0.5` .
- Therefore, maximum power transfer occurs when `delta = 90°` .

- Substituting `δ* = 0.5` into the power equation gives: `P_average_maximum = V_in^2 / (8 * N * L * FSW)`
- The power transfer characteristic is parabolic, starting from zero at `δ* = 0` and `δ* = 1`, and peaking at `δ* = 0.5`.
- **Inductance (L) Selection:**
- The maximum value of `L` is constrained by the requirement to transfer the maximum desired power (`P_average_maximum`) even at minimum input and output voltages (`V_in_min` , `V_out_min`).
- `L <= (V_in_min * V_out_min) / (8 * N * FSW * P_average_maximum)`
- `FSW` (switching frequency) and `N` (turns ratio) are designer's choices, influencing losses and component size. Iterative design is often needed to optimize `L` for minimal losses and maximum power density.
- The `delta_star` range is typically clamped (e.g., 0.05 to 0.95) in microcontrollers, which also influences the minimum `L` required for a given power transfer.
- **Inductor Current Waveform (V_in < V_out/N):**
- When `V_in` is less than `V_out/N` , the inductor current waveform changes significantly.
- Instead of rising then rising, then falling then falling, the waveform typically rises, then falls, then falls, then rises.
- This change in waveform shape impacts the RMS current and needs to be considered during design. The turns ratio `N` can be adjusted to ensure `V_in` is either consistently greater or less than `V_out/N` to optimize RMS current.

3. Backward Power Flow

Bidirectional power flow is a key advantage of DAB converters.

- **Mechanism:** To achieve backward power flow (from A2B2 to A1B1), a leading phase shift must be applied to `V_A2B2` with respect to `V_A1B1` .
- **Waveform Impact:** The inductor current waveform will again change, reflecting the reversed voltage difference across the inductor. The power equation will show a negative sign, indicating power flowing out of the primary side.
- The magnitude of backward power can be controlled by the phase shift, similar to forward power flow.

4. Impact of Modulation on RMS Current

The RMS current flowing through the inductor, transformer windings, and switches is a critical parameter as it directly determines the conduction losses and thermal requirements of the converter.

- **RMS Current and Losses:** Higher RMS currents lead to increased `I^2R` losses in the inductor, transformer windings, and the ON-state resistance of the switches.
- **Modulation Scheme Influence:** Different modulation schemes (SPS, DPS, TPS) result in different inductor current waveforms, and thus different RMS current values for the same amount of power transfer.
- **Optimization:** Designers often perform optimization to select the modulation scheme and converter parameters (like `N` and `L`) that minimize the RMS current for the desired operating range, thereby maximizing efficiency. This involves analyzing the complex `IL` waveforms generated by DPS and TPS, which often have more segments and zero-current intervals compared to SPS.

5. Advanced DAB Topologies

Beyond the basic DAB, several derivatives exist to address specific performance requirements.

- **Resonant DAB Converters:**
- Instead of applying square wave voltages directly to the transformer, resonant DABs use resonant networks (e.g., series L-C circuits) to shape the voltage waveforms into quasi-sinusoidal forms.
- This allows for soft-switching (Zero Voltage Switching - ZVS or Zero Current Switching - ZCS) of the switches, significantly reducing switching losses.
- The resonant network also filters out higher harmonics, ensuring that primarily the fundamental switching frequency component is applied to the transformer, improving efficiency and reducing EMI.
- **Multi-level DAB Converters:**
- These converters replace the standard two-level half-bridges with multi-level half-bridges (e.g., three-level or five-level).
- By generating more voltage levels, the voltage stress on individual switches is reduced, and the voltage waveforms applied to the inductor and transformer are closer to sinusoidal.

- This leads to lower RMS currents, reduced harmonic content, and improved overall efficiency and power quality, especially for higher power applications.
- Modular multi-level DABs further extend this concept for very high power ratings.

Key Takeaways

- [ ] Dual Active Bridge (DAB) converters enable efficient, isolated, and bidirectional power transfer, crucial for EV charging.
- [ ] Single Phase Shift (SPS) modulation is the simplest control method, using a single phase shift `delta` to regulate power.
- [ ] Maximum power transfer in SPS modulation occurs when the normalized phase shift `delta_star` is 0.5 (i.e., `delta = 90°`).
- [ ] The maximum power transfer capability is inversely proportional to the inductance `L` and directly proportional to the square of the input voltage and switching frequency.
- [ ] Inductance `L` must be carefully selected to ensure the desired maximum power can be transferred at minimum voltage conditions, while also considering RMS current.
- [ ] Backward power flow is achieved by applying a leading phase shift to the secondary bridge voltage relative to the primary.
- [ ] Dual and Triple Phase Shift (DPS, TPS) modulations offer more control degrees of freedom, potentially reducing RMS currents and improving efficiency compared to SPS.
- [ ] RMS current in the inductor, transformer, and switches is a critical factor determining conduction losses and overall converter efficiency.
- [ ] Advanced DAB topologies like Resonant DABs and Multi-level DABs improve performance by reducing switching losses and RMS currents, respectively.

Conclusion

The Dual Active Bridge converter is a cornerstone technology in modern power electronics, particularly for applications demanding high efficiency, galvanic isolation, and bidirectional power flow, such as EV charging infrastructure. The detailed analysis of its modulation schemes, inductor current dynamics, and power transfer capabilities provides a robust framework for designing and optimizing these converters. By understanding how to manipulate phase shifts and select appropriate inductance values, engineers can tailor DAB converters to meet specific

performance targets, whether it's maximizing power density, minimizing losses, or enabling advanced functionalities like vehicle-to-grid (V2G) energy exchange. As we move forward, the principles discussed here will serve as a foundation for exploring further design considerations, including the critical role of transformer design and the integration of communication aspects in comprehensive EV charging systems.